EVA-Flow: Learning Shared Chemistry for Unified Environment-Aware Molecular Conformation Generation

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Abstract

Molecular 3D structures underpin simulations of thermodynamics, kinetics, crystallization, and nucleation, making their accurate prediction central to drug discovery and materials design. However, molecular geometry can change dramatically across environments. Existing models either ignore environmental effects or train separate models for each environment. We hypothesize that transfer learning across contexts is possible for conformer generation and that a general model will perform better when training data are limited, as in ligand binding and crystallization. We introduce *EVA-Flow*, a unified framework for environment-aware molecular conformation generation that learns shared chemical structure and adapts it through environment-conditioned generative modeling. Across vacuum, protein-ligand docking, solvation, and crystal packing, pretraining and cross-environment finetuning provide their largest gains in the three data-scarce settings. Evaluation against multiple reference conformations shows that EVA-Flow covers a broad range of solvation conformations. Analyses of molecules shared across environments support its ability to generate different conformations for the same molecule in different environments.

Code release: `https://huggingface.co/facebook/eva-flow`.

1 Introduction

Molecules do not exist in a single fixed shape. A drug molecule may be synthesized in solution, crystallized for storage, redissolved in the body, and finally bind a target protein. At each stage the surrounding environment, whether solvent (Sobornova et al., 2024), crystal lattice (Chattopadhyay et al., 2025; Thompson & Day, 2014; Cruz-Cabeza & Bernstein, 2014), or protein pocket (Gao et al., 2010; Wang & Pang, 2007; Weikl & Paul, 2014), reshapes its 3D geometry.

Despite this, most generative models treat conformer generation as environment-agnostic: molecules are modeled in vacuum (Xu et al., 2022; Hassan et al., 2024) or within a single context such as protein–ligand docking (Corso et al., 2022; 2024), with a separate model for each setting. This fragmentation implicitly

assumes that conformational distributions across environments are unrelated tasks. Yet while conformations change across environments, the underlying chemistry does not—the environment imposes physical forces that deform a shared molecular structure into different geometries.

This motivates formulating conformer generation as learning *shared invariant chemistry* under environment-dependent physical constraints, enabling a single model to transfer chemical knowledge across environments via environment-specific conditioning.

We propose *EVA-Flow*, a unified generative framework for environment-aware molecular conformation generation (Figure 1). EVA-Flow couples a variational autoencoder (VAE) (Kingma et al., 2019) encoder with a flow-matching (FM) decoder (Lipman et al., 2022) that generates conformations conditioned on the molecular graph G and environment embedding e.

Our contributions are:

- We formulate conformer generation as learning shared chemistry under environment-specific constraints, rather than training isolated models per environment.
- We introduce a unified environment-conditioned flow-matching architecture that handles diverse environments within a single model, combining an environment encoder, a learned harmonic base, and a variational latent that together support low-data cross-environment transfer.
- We demonstrate that pretraining and cross-environment finetuning improve accuracy most clearly in data-scarce environments, and provide same-molecule and multi-reference evidence that the model responds to environmental context. UMA relaxation separately supports physical validity for generated Vacuum conformers (Appendix F).

2 Related Work

Flow matching. Flow matching (FM) trains a continuous normalizing flow by regressing the velocity field that transports a base distribution to the data, yielding a simulation-free objective (Lipman et al., 2022). Rectified flow (Liu et al., 2022) and conditional FM (Tong et al., 2023) extend this to straight-line paths and conditional generation. Equivariant (Hassan et al., 2024) and geometric (Chen & Lipman, 2023) variants adapt FM to 3D molecular settings.

VAE + FM. Hybrid VAE–flow models take two forms: (1) FM in the latent space replaces the VAE prior (Dao et al., 2023), or (2) FM serves as the decoder, mapping a base distribution to data conditioned on a latent code (Fischer et al., 2023; Sargent et al., 2025). We adopt the latter, using an environment-aware FM decoder for conformer generation.

Molecular conformation generation. Classical force-field and MD approaches are accurate but expensive. Neural methods generate conformers via equivariant diffusion (Xu et al., 2022), flows (Hassan et al., 2024), or torsion models (Jing et al., 2022). Most target a single environment—typically vacuum or protein–ligand docking (Corso et al., 2024)—requiring separate models per setting. EVA-Flow addresses this limitation with a single model that conditions on structured environments.

Diffusion in latent space for molecules. Joshi et al. (2025) run diffusion in a learned latent space and decode to coordinates. EVA-Flow also uses a VAE encoder but performs FM in data space and explicitly conditions on the environment.

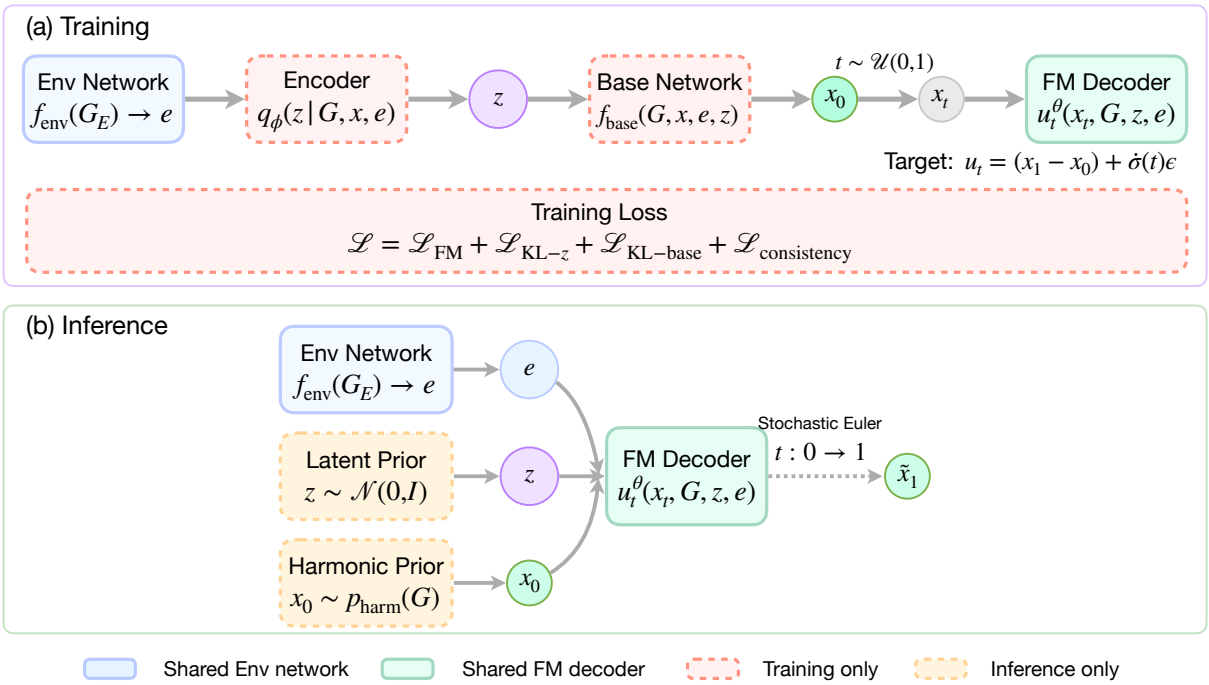


Figure 1: Architecture of EVA-Flow. **(a) Training.** The environment graph G_E is embedded by f_{env} into a vector e. The encoder $q_\phi(z \mid G, x, e)$ maps the molecular graph G, ground-truth positions x, and environment e to a latent z. The base network $f_{\text{base}}(G, x, e, z)$ produces a distribution $p_0(x_0 \mid G, x, e, z)$ from which x_0 is sampled. The FM decoder learns $u_t^\theta(x_t, G, z, e)$ with stochastic interpolation $x_t = (1-t)\, x_0 + t\, x_1 + \sigma(t)\epsilon$, targeting $u_t = (x_1 - x_0) + \dot{\sigma}(t)\epsilon$. Total loss: $\mathcal{L} = \mathcal{L}_{\text{FM}} + \mathcal{L}_{\text{KL-z}} + \mathcal{L}_{\text{KL-base}} + \mathcal{L}_{\text{consistency}}$. **(b) Inference.** The encoder and base network are bypassed: $e = f_{\text{env}}(G_E)$, $z \sim \mathcal{N}(\mathbf{0}, I)$, and $x_0 \sim p_{\text{harm}}(G)$ are sampled independently. All three condition the FM decoder, which generates conformations from t=0 to t=1 using the stochastic Euler sampler described in Section 3.2. The Env Network and FM Decoder share weights between training and inference.

3 Environment-Aware Flow Matching

3.1 Problem Formulation

Let $G = (V, \mathcal{E})$ denote a molecular graph with nodes representing atoms and edges representing bonds. Each atom is associated with a 3D position $x \in \mathbb{R}^{3N}$, where N is the number of atoms. A molecule exists in an environment E, such as a protein pocket, solvent box, or crystalline lattice. Our general objective is the conditional distribution $p(x|G, E)$, capturing how environments modulate molecular conformations. We seek one model that learns shared molecular structure and transfers across the environment types observed during training.

EVA-Flow uses two graph objects with distinct roles. The *target molecular graph* G always describes the molecule whose centered coordinates x are generated; its nodes are molecular atoms and its edges are covalent bonds. The *environment graph* G_{E} is conditioning context only: it is embedded into a fixed-dimensional vector $e = f_{\text{env}}(G_{\text{E}})$ and is never generated. Table 1 makes the environment-specific constructions explicit. During training the encoder and learned base additionally access the target coordinates x; at inference only G, G_{E}, and their listed attributes are required.

The molecular encoder/decoder backbone and conditioning mechanism are shared across environments. Environment-specific graph constructions incorporate domain knowledge (Table 1).

Table 1: Roles and construction of the target graph G and environment graph G_{E}. The target graph has the same semantics in every setting; only the context representation changes. "Coordinates" indicates coordinates supplied as input, not the target coordinates to be generated.

Stage/ environment	Target G	G_{E} nodes	G_{E} edges	Context attributes / coordinates	Available at inference
Pretraining	Molecule atoms/bonds	Random 30–70% atom-induced subgraph	Retained covalent bonds	Atom features and masked-subgraph coordinates	N/A (pretraining only)
Vacuum	Molecule atoms/bonds	Bemis–Murcko scaffold atoms	Scaffold covalent bonds	Atom features; coordinates set to zero	Yes, constructed from input molecule
Docking	Ligand atoms/bonds	Protein-pocket heavy atoms	Atoms within the same residue	Atom features and pocket coordinates	Yes, extracted pocket
Solvation	Solute atoms/bonds	Atoms of a water proxy	Water covalent bonds	sCAV, vCAV, eSOLV, eNELEC; fixed proxy coordinates	Yes, solvent descriptors
Packing	Molecule atoms/bonds	Copy of molecular atoms	Molecular covalent bonds	RDKit geometry, space group, lattice $(a, b, c, \alpha, \beta, \gamma)$	Yes, crystal metadata

3.2 Model Architecture

EVA-Flow is a VAE with an FM decoder (Figure 1).

1. **Environment Embedding Network** (Figure 1). The environment is represented as a graph $G_{\mathrm{E}} = (V_{\mathrm{E}}, \mathcal{E}_{\mathrm{E}})$, where nodes correspond to atoms with associated features and edges correspond to bonds or interactions. The construction of G_{E} is described in Section 4.1. The environment network, f_{env}, consists of three graph convolutional network (GCN) layers (Zhang et al., 2019), followed by a two-layer MLP projection to the environment embedding vector $e \in \mathbb{R}^d$:

$$e = f_{\mathrm{env}}(G_{\mathrm{E}}) \tag{1}$$

   The embedding e conditions both the molecular encoder and decoder. The GCN concatenates coordinates with node features and is not rotation invariant. Appendix O compares it with an SE(3)-equivariant environment encoder.

2. **Encoder** (Figure 1). The encoder is a GNN-based module that takes as input the molecular graph G, atomic positions x, and the environment embedding e. Input node features are formed by concatenating the atomic number, node attributes, and atomic positions. These features are processed by an SE(3)-equivariant transformer (TorchMD-ET) (Thölke & Fabritiis, 2022; Hassan et al., 2024), producing node embeddings of size [N, hidden]. Each node embedding is then concatenated with the environment embedding e, and projected through two linear layers to produce the mean and log-variance for a node-level latent variational posterior distribution:

$$\begin{aligned} & q_\phi(z \mid G, x, e) \\ &= \prod_{i=1}^{N} \mathcal{N}\Big(z_i \,\Big|\, \mu_{\phi,i}(G, x, e), \mathrm{diag}\big(\sigma^2_{\phi,i}(G, x, e)\big)\Big) \\ &= \mathcal{N}\big(\mu_\phi(G, x, e),\ \mathrm{diag}(\sigma^2_\phi(G, x, e))\big). \end{aligned} \tag{2}$$

   where $\mu_{\phi,i}, \sigma^2_{\phi,i} \in \mathbb{R}^d$ denote the mean and variance predicted for node i. Stacking these predictions over all nodes gives $\mu_\phi, \sigma^2_\phi \in \mathbb{R}^{N \times d}$. The latent pathway captures residual molecule–environment variability during training. The reported checkpoint has a collapsed latent posterior, while the coupled base and latent aid low-data cross-environment transfer (Appendices M and J).

3. **Base Distribution Network** (Figure 1). We use an SE(3)-equivariant transformer (TorchMD-ET) to parameterize a conditional Harmonic base distribution over atomic coordinates. The base distribution is

commonly parameterized with zero mean and the target molecule's graph Laplacian as precision matrix. We condition this distribution by predicting the mean and diagonal scaling:

$$\mu_{\text{base}},\, \lambda_{\text{base}} = f_{\text{base}}(G, x, e, z), \tag{3}$$

where $\mu_{\text{base}} \in \mathbb{R}^{N\times 3}$ is vector-valued and $\lambda_{\text{base}} \in \mathbb{R}^N$ is scalar-valued. Our conditional distribution is then:

$$L_{\text{base}} = L + \text{diag}(\lambda_{\text{base}}), \tag{4}$$

$$\Sigma_{\text{base}} = L_{\text{base}}^{\dagger}, \tag{5}$$

$$p_0(x_0 \mid G, x, e, z) = \mathcal{N}\Big(x_0 \,\Big|\, \mu_{\text{base}}, \Sigma_{\text{base}}\Big). \tag{6}$$

where $(\cdot)^{\dagger}$ is the Moore–Penrose pseudoinverse to handle global translational symmetries. Let $L_{\text{base}} = PDP^{\top}$ be the eigendecomposition. Sampling is performed via:

$$x_0 = \mu_{\text{base}} + PD^{-1/2}\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I) \tag{7}$$

The network is trained by minimizing the KL divergence between the learned distribution and a Harmonic prior defined by the molecular graph topology $\mathcal{N}(\mathbf{0}, L^{\dagger})$. During training the network accesses the ground truth x for conditioning. The base network is not used at inference.

4. **FM Decoder** (Figure 1). The decoder generates conformations by transporting samples from the base distribution p_0 to the data distribution using stochastic flow matching. We adopt the TorchMD equivariant network to parameterize a time-dependent vector field u_t^{θ} that takes as input the current molecular structure x_t, the molecular graph G, the latent variables z, and the environment embedding e:

$$x_0 \sim p_0'(x_0) = \begin{cases} p_0(x_0 \mid G, x, z, e) & \text{Training} \\ \mathcal{N}\left(x_0 \mid \mathbf{0}, L^{\dagger}\right) & \text{Inference} \end{cases} \tag{8}$$

$$\dot{x}_t = u_t^{\theta}(x_t, G, z, e) \tag{9}$$

We use a stochastic interpolation between base and target samples. Given a base sample $x_0 \sim p_0'$ and a data sample $x_1 \sim p_{\text{data}}$, we define:

$$\begin{aligned} \mu_t &= (1-t)x_0 + tx_1, \\ x_t &= \mu_t + \sigma(t)\,\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I) \end{aligned} \tag{10}$$

We use the noise schedule:

$$\sigma(t) = \sigma\sqrt{t(1-t)} \tag{11}$$

5. **Inference** (Figure 1). At inference time, given the environment graph G_E, we compute the embedding $e = f_{\text{env}}(G_E)$ and sample a latent $z \sim p(z) = \mathcal{N}(\mathbf{0}, I)$. The initial positions are sampled from the simple Harmonic prior $x_0 \sim \mathcal{N}(\mathbf{0}, L^{\dagger})$. Starting at $x_{t_0} = x_0$, we integrate the learned flow field from $t = 0$ to $t = 1$ using stochastic Euler steps with zero-center-of-mass Gaussian perturbations. For $0 = t_0 < t_1 < \cdots < t_n = 1$,

$$\begin{aligned} x_{t_{i+1}} = x_{t_i} &+ u_{\theta}(x_{t_i}, t_i; G, z, e)\,\Delta t_i \\ &+ \sqrt{\sigma_{\text{samp}}^2(t_{i+1}) - \sigma_{\text{samp}}^2(t_i)}\,\xi_i, \\ \xi_i \sim \mathcal{N}(\mathbf{0}, I), &\qquad \text{with COM}(\xi_i) = 0, \end{aligned} \tag{12}$$

where the sampler uses the variance schedule $\sigma_{\text{samp}}^2(t) = t^2$ and $\Delta t_i = t_{i+1} - t_i$. We refer to this procedure as the "SDE sampler" in the experimental tables (Appendix C.1). The final conformer $\hat{x}$ is obtained after n steps.

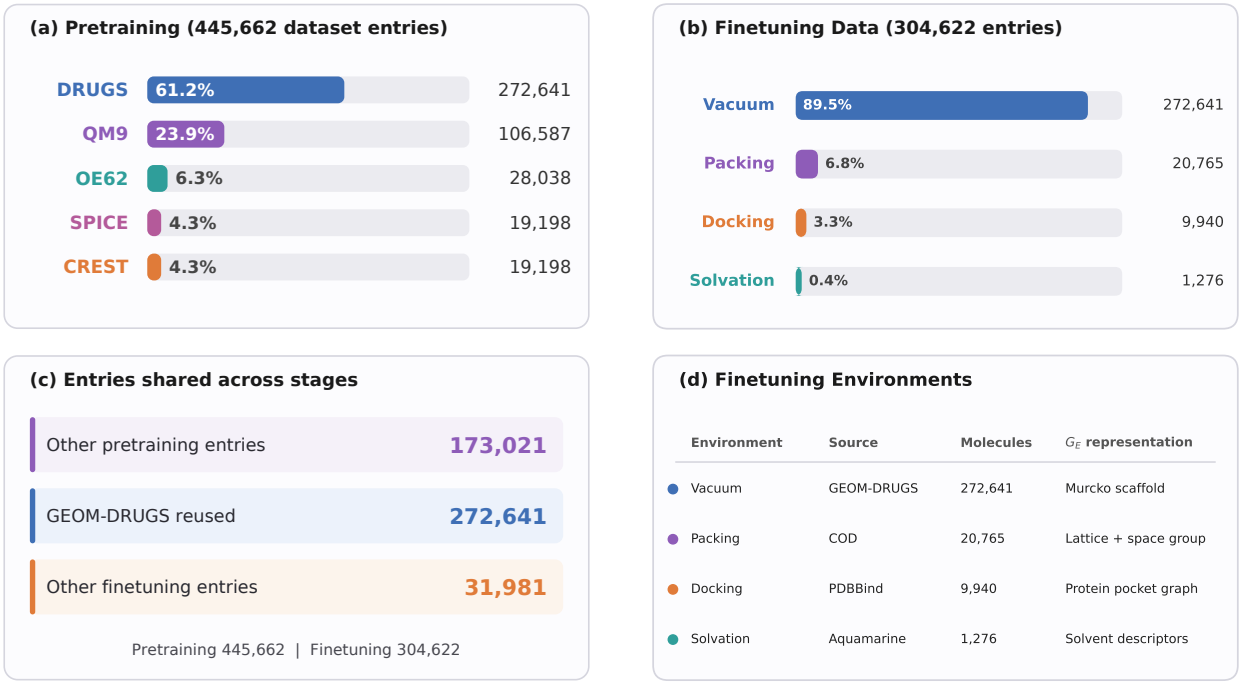


Figure 2: Dataset composition. **(a)** Pretraining corpus by source (445,662 molecule entries), counting SPICE and CREST-relaxed SPICE separately. **(b)** Finetuning corpus by environment (304,622 molecule entries). **(c)** Reuse of all 272,641 GEOM-DRUGS entries across pretraining and Vacuum finetuning; the remaining counts denote entries in the other datasets. **(d)** Data source and environment graph G_E representation for each finetuning environment.

3.3 Training Objectives

We jointly optimize the encoder, environment embedding network, base distribution network, and FM decoder with four loss terms (Figure 1):

1. **Latent Prior Regularization**. This VAE loss regularizes the latent posterior toward a standard normal distribution:
$$\mathcal{L}_{\mathrm{KL-z}} = D_{\mathrm{KL}}(q_\phi(z|x,G,e)\,||\,\mathcal{N}(\mathbf{0},I)). \tag{13}$$

2. **Base Distribution Regularization**. The learned base distribution is regularized by KL divergence toward a simple Harmonic prior induced by graph topology L:
$$\mathcal{L}_{\mathrm{base}} = D_{\mathrm{KL}}\big(\mathcal{N}(\mu_{\mathrm{base}},\Sigma_{\mathrm{base}})\,\|\,\mathcal{N}(\mathbf{0},L^\dagger)\big). \tag{14}$$

3. **Latent Consistency Loss**. This loss penalizes disagreement between the encoder's posterior mean given target conformation x and initial conformation $x_0 \sim p_0'(x_0)$:
$$\mathcal{L}_{\mathrm{consistency}} = ||\mu_\phi(x_0,G,e) - \mu_\phi(x,G,e)||_2^2. \tag{15}$$

4. **FM Loss**. This loss trains the decoder to transport samples from the base distribution to the data distribution through continuous-time dynamics.
$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,x_t}[||u_t^\theta(x_t,G,z,e) - u_t(x_0,x)||^2], \tag{16}$$
where u_t is the analytically defined target velocity field:
$$\begin{aligned} u_t &= (x_1 - x_0) + \dot\sigma(t)\,\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0},I), \\ \dot\sigma(t) &= \sigma\frac{1-2t}{2\sqrt{t(1-t)}} \end{aligned} \tag{17}$$

All networks are trained jointly at each iteration. We weight all loss terms equally:

$$\mathcal{L} = \mathcal{L}_{\mathrm{KL-z}} + \mathcal{L}_{\mathrm{KL-base}} + \mathcal{L}_{\mathrm{consistency}} + \mathcal{L}_{\mathrm{FM}}. \tag{18}$$

These terms regulate two train–inference transitions while preserving coordinate accuracy. $\mathcal{L}_{\mathrm{FM}}$ is the primary geometric objective and learns the coordinate transport. $\mathcal{L}_{\mathrm{KL-z}}$ makes sampling $z \sim \mathcal{N}(\mathbf{0}, I)$ valid at inference; $\mathcal{L}_{\mathrm{KL-base}}$ similarly aligns the learned training-time base with the graph-only harmonic prior used at inference. The consistency term prevents the latent from encoding incidental coordinate-level differences by requiring the same molecule–environment state to have a stable representation at x and x_0. Thus, the KL and consistency terms constrain information pathways that could otherwise become training-time shortcuts, while the FM term retains the information required for accurate generation.

4 Experiments

4.1 Datasets

Pretraining. For large-scale pretraining, we combine datasets containing conformations of small organic and drug-like molecules. This includes relaxed conformations from GEOM-QM9 and GEOM-DRUGS (Axelrod & Gomez-Bombarelli, 2022; Nikitin et al., 2025), as well as higher-energy conformations from the SPICE PubChem subset (Eastman et al., 2023). We also relax SPICE conformations using `CREST` (Pracht et al., 2024) to obtain low-energy conformations. We include the OE62 dataset (Stuke et al., 2020), which consists of crystal-forming molecules relaxed in vacuum. The pretraining corpus contains 445,662 molecule entries across five datasets and 9.8M conformations (Figure 2a). SPICE and CREST-relaxed SPICE each contribute 19,198 entries for the same molecules.

During pretraining, we construct G_{E} by uniformly sampling a mask ratio between 0.3 and 0.7 and retaining the corresponding atom-induced subgraph of the target molecule, including its covalent edges and coordinates. This denoising-style context exposes the shared backbone to diverse partial molecular environments. During Vacuum finetuning and evaluation, we instead apply the RDKit (Landrum & contributors, 2025) `MurckoScaffold` package (RDKit, 2025) to extract the molecule's Bemis–Murcko scaffold (Bemis & Murcko, 1996); scaffold coordinates are set to zero so that the context encodes topology rather than the target conformation.

Finetuning. We finetune our model in four environments:

1. **Vacuum**. Molecules in vacuum are isolated. We use the GEOM-DRUGS dataset and construct G_{E} from the molecule's Bemis–Murcko scaffold as described above (Figure 2(d)).
2. **Protein–Ligand Docking**. In this task, molecules (ligands) adapt their internal conformation to a known protein binding pocket. We use the PDBBind-v2020 dataset (Liu et al., 2015). The environment graph is constructed using heavy atoms in the extracted protein pocket; edges join atoms in the same residue, and pocket coordinates are included (Figure 2(d)). We center the ligand and predict its pocket-conditioned internal conformation; unlike end-to-end docking systems, EVA-Flow does not predict the ligand's global translation or orientation in the protein.
3. **Solvation**. Dissolved molecules interact with the solvent. We use the Aquamarine dataset (Medrano Sandonas et al., 2024), which includes conformations relaxed in implicit water. To construct the environment graph, we use a water molecule and add solvation descriptors to the node features such as cavity surface area (sCAV), cavity volume (vCAV), free energy in electrolyte (eSOLV), and non-electrostatic free energy (eNELEC) (Figure 2(d)). These descriptors represent the interaction between molecules and solvent.
4. **Crystal Packing**. Molecules interact with neighboring molecules in crystals. We use the Crystallography Open Database (COD) (Gražulis et al., 2009; 2012) and retain only small organic molecular crystals. We convert the fractional coordinates from the CIF files to Cartesian positions, and recover the bonds with `OpenBabel` (O'Boyle et al., 2008; 2011). For the environment graph, we use a copy of the molecular graph with an RDKit proxy geometry and append the space-group number (sg) and lattice parameters $(a, b, c, \alpha, \beta, \gamma)$ to its features (Figure 2(d)).

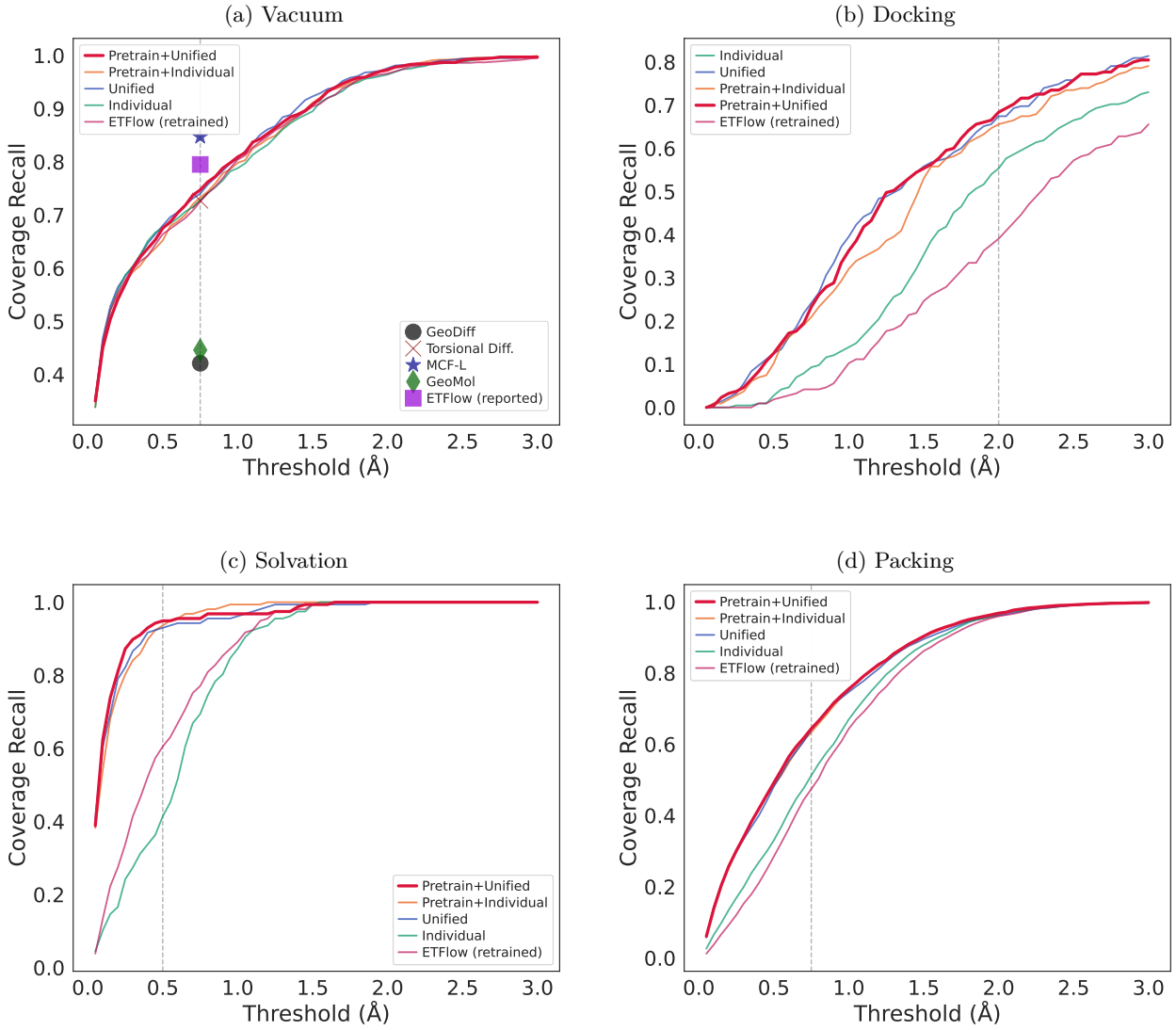


Figure 3: Recall COV as a function of the RMSD threshold across the four environments. In every panel the four EVA-Flow training setups (Individual, Unified, Pretrain+Individual, Pretrain+Unified) are compared to retrained ETFlow. **(a)** Vacuum (cleaned GEOM-DRUGS (Nikitin et al., 2025)); reference points denote published Recall COV from Xu et al. (2022); Ganea et al. (2021); Jing et al. (2022); Wang et al. (2024); Hassan et al. (2024). **(b)** Docking (PDBBind), **(c)** Solvation (Aquamarine), and **(d)** Packing (COD) compare models trained and evaluated under the same centered-conformation protocol. Dashed vertical lines mark the environment-specific RMSD thresholds (0.75/2.0/0.5/0.75 Å).

Sections A and B describe data processing and the distributions of atom and rotatable-bond counts.

4.2 Experimental Setup

We compare four training strategies.

1. **Pretraining + Individual Finetuning**. We pretrain the model on the pretraining datasets to learn shared chemistry from many molecules and conformations, then finetune it on a single environment.

2. **Pretraining + Unified Finetuning**. We pretrain the model on the datasets in Section 4.1, then finetune it on all finetuning environments. This setup tests adaptation across environments.

3. **Unified Finetuning**. We train the model across all environments without pretraining.

4. **Individual Finetuning**. We train separate models for each environment without pretraining. This provides a baseline for assessing the effects of unification and pretraining.

Evaluation protocols. The available supervision differs by benchmark: Vacuum and Solvation provide multi-conformer references for ensemble evaluation, whereas Docking and Packing provide one physical reference and therefore test environment-conditioned structure prediction. The **4-conformer protocol** is our default across all four environments: each model generates 4 conformers per test molecule, evaluated against one reference (Table 2, Figure 3). We report mean COV and mean/median AMR. The **2K protocol** generates $2K$ conformers for a molecule with K references. We use it for Vacuum comparisons with published baselines (Ganea et al., 2021; Jing et al., 2022; Wang et al., 2024; Hassan et al., 2024) (Table 3) and multi-reference Solvation evaluation against all available AQM conformers (Appendix E.4).

Evaluation metrics. We evaluate generated conformers against ground-truth conformers using distance-based RMSD. We report the Recall and Precision Coverage (COV, ↑) and average minimum RMSD (AMR, ↓) (Section D). For crystal packing, we also align the generated conformation with the ground-truth one and compute a symmetry-aware RMSD (sRMSD). This value reflects the minimal atomic displacement after accounting for lattice symmetry and periodicity.

4.3 Results

Table 2 and Figure 3 summarize results under the default 4-conformer protocol. On Vacuum under the 2K protocol, EVA-Flow is comparable to retrained ETFlow and has lower Recall COV but higher Precision COV than MCF-L (Table 3). Its accuracy gains are concentrated in the data-scarce Docking, Solvation, and Packing settings.

Pretraining improves generalization. Pretraining provides gains over Individual training in the data-scarce environments (Table 2, Figure 3). Vacuum differences among EVA-Flow setups are small relative to the observed inference-sampler variability, consistent with its much larger training set.

Unified training outperforms individual models. Unified setups substantially surpass Individual finetuning in Docking, Solvation, and Packing, whereas the differences within Vacuum are small. A data-volume control shows that the gains persist beyond raw target-environment exposure (Appendix K).

Baseline comparisons. We compare with retrained ETFlow using matched task definitions and evaluation code. EVA-Flow also outperforms ETKDG+MMFF in all four environments and GFN2-xTB on Solvation, where both physical references collapse (Appendix E.5). Baseline training details and comparison scope are provided in Appendix E.6.

4.4 Analysis

We evaluate environment conditioning using molecules matched by canonical SMILES across all six environment pairs. For each pair, the Pretrain+Unified model generates conformations of the same molecules conditioned on each environment. Figure 4 and Table 5 compare these generations with the environment-specific references using mean heavy-atom RMSD after alignment.

Across all six environment pairs, generated conformations are closer on average to their own environment's reference than to the other environment's reference (Table 5). The gap is largest for Solvation–Vacuum and smallest for Packing–Docking. Together with the visualizations in Table 4, this pattern supports EVA-Flow's ability to generate environment-specific conformations for the same molecule.

Table 2: Conformation generation under the **4-conformer protocol** (default) across four environments. Each model generates 4 conformers per molecule, compared with one reference. We report Recall and Precision COV (↑, mean) and AMR (↓, mean and median) at environment-specific RMSD thresholds: 0.5 Å for Solvation, 0.75 Å for Vacuum and Packing, and 2.0 Å for Docking. For Packing, we additionally report symmetry-aware RMSD (sRMSD, ↓, mean). Except for sRMSD, metric cells are mean ± standard deviation over five independent SDE sampling seeds from one fixed trained checkpoint (Appendix C.1). Best means within each environment are **bolded**.

Environment	Setup	Recall			Precision			sRMSD↓
		COV↑	AMR↓		COV↑	AMR↓		
		mean	mean	median	mean	mean	median	
Vacuum	Pretrain+Unified	**0.747 ±0.013**	0.468 ±0.012	0.144 ±0.024	0.463 ±0.004	0.969 ±0.006	0.900 ±0.023	–
	Pretrain+Individual	0.733 ±0.010	0.481 ±0.012	0.146 ±0.019	0.454 ±0.004	0.980 ±0.008	0.925 ±0.016	–
	Unified	0.740 ±0.007	**0.456 ±0.008**	0.131 ±0.011	0.463 ±0.003	**0.954 ±0.005**	**0.879 ±0.011**	–
	Individual	0.728 ±0.005	0.487 ±0.004	0.134 ±0.006	0.460 ±0.003	0.978 ±0.004	0.911 ±0.017	–
	ETFlow (retrained)	0.725 ±0.012	0.488 ±0.011	**0.130 ±0.013**	**0.467 ±0.007**	0.972 ±0.013	0.919 ±0.022	–
Docking	Pretrain+Unified	**0.684 ±0.013**	1.899 ±0.027	**1.257 ±0.037**	0.512 ±0.008	2.471 ±0.023	1.864 ±0.039	–
	Pretrain+Individual	0.656 ±0.010	1.980 ±0.013	1.463 ±0.021	0.458 ±0.006	2.581 ±0.019	2.076 ±0.034	–
	Unified	0.674 ±0.021	**1.889 ±0.016**	1.303 ±0.060	**0.536 ±0.012**	**2.391 ±0.021**	**1.732 ±0.023**	–
	Individual	0.553 ±0.013	2.458 ±0.034	1.833 ±0.039	0.378 ±0.009	3.135 ±0.032	2.387 ±0.075	–
	ETFlow (retrained)	0.391 ±0.016	2.768 ±0.029	2.288 ±0.084	0.242 ±0.012	3.580 ±0.019	2.958 ±0.052	–
Solvation	Pretrain+Unified	**0.949 ±0.019**	**0.153 ±0.013**	**0.072 ±0.016**	**0.780 ±0.009**	**0.367 ±0.007**	**0.232 ±0.013**	–
	Pretrain+Individual	0.936 ±0.018	0.157 ±0.006	0.087 ±0.020	0.693 ±0.006	0.431 ±0.007	0.334 ±0.017	–
	Unified	0.930 ±0.034	0.166 ±0.017	0.083 ±0.018	0.769 ±0.014	0.370 ±0.009	0.249 ±0.023	–
	Individual	0.414 ±0.021	0.580 ±0.020	0.568 ±0.026	0.177 ±0.009	0.961 ±0.045	0.851 ±0.044	–
	ETFlow (retrained)	0.605 ±0.034	0.475 ±0.025	0.366 ±0.035	0.264 ±0.013	0.824 ±0.015	0.786 ±0.028	–
Packing	Pretrain+Unified	**0.644 ±0.004**	**0.664 ±0.002**	**0.509 ±0.005**	0.410 ±0.001	1.073 ±0.001	1.008 ±0.005	0.055
	Pretrain+Individual	0.633 ±0.005	0.672 ±0.003	0.529 ±0.006	0.400 ±0.002	1.087 ±0.001	1.022 ±0.005	0.055
	Unified	0.640 ±0.003	0.682 ±0.005	0.527 ±0.006	**0.414 ±0.002**	**1.067 ±0.002**	**0.997 ±0.008**	**0.054**
	Individual	0.512 ±0.006	0.819 ±0.004	0.734 ±0.008	0.303 ±0.002	1.222 ±0.004	1.161 ±0.004	0.064
	ETFlow (retrained)	0.474 ±0.003	0.878 ±0.005	0.794 ±0.006	0.274 ±0.002	1.274 ±0.006	1.203 ±0.009	0.069

Packing–Docking has similar reference geometries and similar generated conformations across environments, consistent with the model responding to context rather than forcing differences from environment labels. Zeroing domain-specific context features degrades performance in Docking, Solvation, and Packing (Appendix H), confirming the use of environment representations. Generated samples also exhibit non-trivial conformational spread (Appendix E.3).

Table 3: Conformation generation for Vacuum under the **2K protocol**: each model generates $2K$ conformers per molecule, where K is the number of reference conformers. We report mean and median Recall and Precision COV (↑) and AMR (↓) at RMSD threshold 0.75 Å. Best results are **bolded**. The models are trained on the original GEOM-DRUGS dataset and evaluated on its full test set. Literature numbers for GeoDiff, GeoMol, Torsional Diffusion, MCF-L, and ETFlow (reported) are taken from Table 1 of Hassan et al. (2024); ETFlow (reported) refers to the ET-Flow base (8.3M) variant. ETFlow (retrained) and EVA-Flow are trained on all available conformers per molecule on the original GEOM-DRUGS and evaluated on the same test set.

Setup	Recall				Precision			
	COV↑		AMR↓		COV↑		AMR↓	
	mean	median	mean	median	mean	median	mean	median
GeoDiff	0.421	0.378	0.835	0.809	0.249	0.145	1.136	1.090
GeoMol	0.446	0.414	0.875	0.834	0.430	0.364	0.928	0.841
Torsional Diff.	0.727	0.800	0.582	0.565	0.552	0.569	0.778	0.729
MCF-L	**0.847**	**0.922**	**0.390**	**0.247**	0.668	0.713	0.618	0.530
ETFlow (reported)	0.795	0.846	0.452	0.419	**0.744**	**0.810**	**0.541**	**0.470**
ETFlow (retrained)	0.712	0.742	0.562	0.530	0.704	0.758	0.594	0.520
EVA-Flow (Pretrain+Unified)	0.721	0.756	0.550	0.527	0.707	0.781	0.589	0.510

Table 4: Reference and generated conformations (Pretrain+Unified) for molecules shared between Solvation and Packing or Solvation and Docking. The conformers are aligned to the Solvation reference.

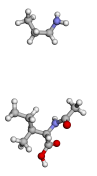
Solvation-Packing				Solvation-Docking			
ref_solv	gen_solv	ref_pack	gen_pack	ref_solv	gen_solv	ref_dock	gen_dock
							

Table 5: Shared-molecule analysis (Pretrain+Unified), corresponding to Figure 4. All values are mean heavy-atom RMSD in Å.

Pair (A/B)	n	own A	own B	cross A	cross B	generation shift
Solvation–Vacuum	142	0.12	0.64	3.41	3.31	3.41
Docking–Vacuum	558	0.75	0.76	1.22	1.21	1.20
Vacuum–Packing	487	0.33	0.35	0.64	0.65	0.64
Docking–Solvation	13	0.55	0.09	2.41	2.49	2.42
Packing–Solvation	23	0.06	0.09	1.49	1.48	1.48
Packing–Docking	64	0.28	0.36	0.45	0.38	0.42

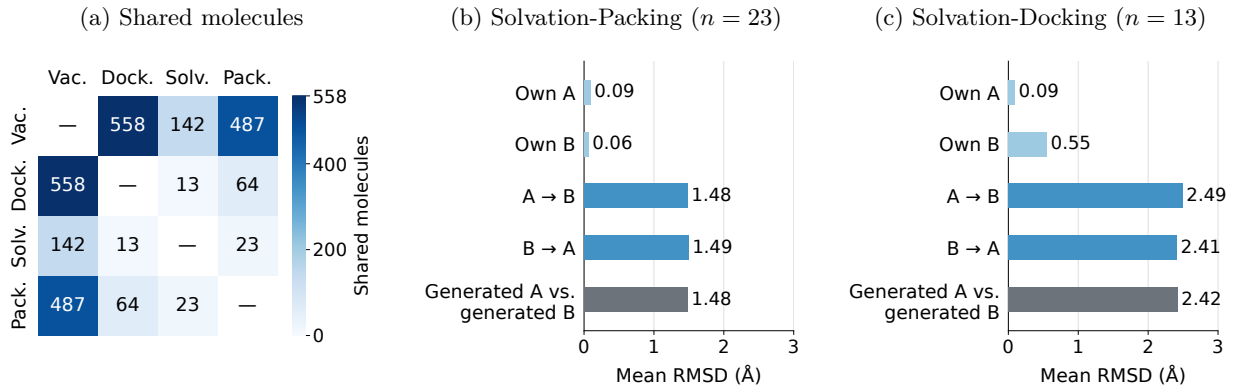


Figure 4: Shared-molecule analysis (Pretrain+Unified), using values from Table 5. (a) Pairwise counts of shared molecules; darker cells indicate more molecules, and dashes mark the diagonal. (b, c) Mean heavy-atom RMSD (Å) after alignment for five comparisons. In both panels, A denotes Solvation; B denotes Packing in (b) and Docking in (c). "Own A" and "Own B" compare each conditioned generation with its own environment's reference. $A \to B$ compares the A-conditioned generation with the reference from B, and conversely for $B \to A$. "Generated A vs. generated B" compares the two environment-conditioned generations and corresponds to generation shift in Table 5.

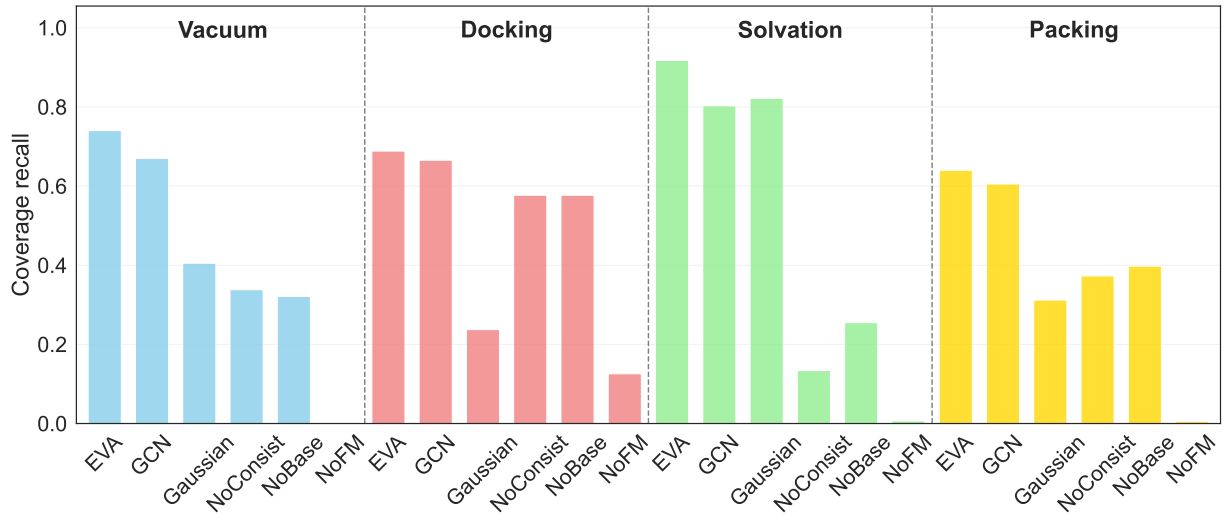


Figure 5: Recall COV for the small model in the Unified setup across Vacuum, Docking, Solvation, and Packing. 'EVA' refers to the standard setup; 'GCN' uses non-equivariant networks for the encoder and base; 'Gaussian' uses a Gaussian base distribution instead of a Harmonic one; 'NoConsist' removes latent consistency loss; 'NoBase' replaces the learned base distribution network for initial positions x_0 with a Harmonic distribution determined only by the graph; 'NoFM' removes the FM decoder and uses a regression decoder.

4.5 Ablation Studies

We ablate components using the Unified setup with the small model (Figure 5). The full model (EVA) is defined in Sections 3.2 and 3.3. Variants: **GCN** replaces SE(3)-equivariant modules with graph convolutional networks; **Gaussian** uses a diagonal Gaussian base instead of the Harmonic distribution; **NoConsist** removes the latent consistency loss; **NoBase** replaces the learned base network with the graph-topology Harmonic prior $\mathcal{N}(\mathbf{0}, L^{\dagger})$; **NoFM** replaces the FM decoder with a regression decoder; and **Latent-free** removes both the learned base and latent, retaining environment conditioning, the fixed harmonic base, and the FM decoder.

Three components are essential across all environments: the FM decoder, the Harmonic base, and SE(3)-equivariance (Figure 5). Replacing the FM decoder with regression collapses Recall COV to 0.000/0.126/0.006/0.005. The Harmonic base outperforms a diagonal Gaussian, and SE(3)-equivariant modules outperform GCN. The learned base and VAE latent provide consistent gains in the tested cross-environment transfer regimes. Removing both lowers Recall COV in all twelve transfer comparisons, with the largest gaps in low-data Solvation (up to -0.255 Recall COV), while effects under Individual training vary by environment (Appendix J). In the Unified ablations, NoBase and NoConsist also reduce Recall COV. Base and latent diagnostics support an optimization role for these components (Appendices L and M). Complete metrics are in Appendix I.

5 Conclusion

We introduced EVA-Flow, a unified framework for environment-aware molecular conformation generation. Its shared flow-matching backbone conditions on heterogeneous representations of vacuum, protein pockets, solvation descriptors, and crystal metadata. The largest gains from pretraining and unified finetuning occur in the three data-scarce environments. Multi-reference Solvation evaluation supports broad ensemble coverage, while shared-molecule and context-ablation analyses support environment-responsive generation.

More expressive backbones and scaling could improve accuracy, while multi-reference benchmarks and environment-specific physical validation would extend evaluation beyond Solvation and Vacuum. Extending to unseen environment types such as surface adsorption or solid–liquid interfaces remains future work.

Broader Impact Statement

EVA-Flow is intended to support molecular modeling in chemistry, materials science, and drug discovery. More accurate conformation generation could reduce the computational cost of downstream simulations. Like other molecular generative and predictive tools, the method may produce inaccurate geometries and should not replace experimental validation or high-accuracy physical simulation in safety-critical applications. We recommend validating generated structures with appropriate physical calculations and domain expertise before downstream use.

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A Data Preparation

GEOM QM9 and DRUGS. We follow the data processing pipeline of ETFlow (Hassan et al., 2024) and adopt the same train/validation/test splits. For GEOM-DRUGS, we applied the data cleaning strategies reported in Nikitin et al. (2025) that fix valency, bond-order, and force-field inconsistencies in the original datasets. For each molecule, we retain the top 30 conformers ranked by Boltzmann weight. Each molecule is stored as a PyTorch Geometric object containing atomic numbers, 3D positions, and a canonical SMILES string. The QM9 training set contains 106,587 molecules (794,960 conformations) and the validation set 13,323 molecules (99,945 conformations); the test set has 1,000 molecules. The DRUGS training set contains 272,641 molecules (5,770,377 conformations) and the validation set 30,405 molecules (715,592 conformations); the test set has 1,000 molecules.

SPICE and CREST-relaxed SPICE. We use the PubChem subset of SPICE (Eastman et al., 2023), which contains small drug-like molecules (18–50 atoms; elements Br, C, Cl, F, H, I, N, O, P, S).

Because the SPICE conformers are not fully relaxed, we construct a complementary relaxed set via CREST (Pracht et al., 2024). For each molecule we (i) generate 50 initial conformers with RDKit `EmbedMultipleConfs` (pruning threshold 0.01 Å, MMFF optimization), (ii) deduplicate with `GetBestRMS` (threshold 0.1 Å), (iii) refine the ten lowest-energy conformers with extended tight binding (xTB) (Grimme et al., 2017; Friede et al., 2024), and (iv) run CREST v3.0.2 from the lowest-xTB-energy conformer with a 6 kcal/mol cutoff. Molecules with cis/trans stereo-SMILES were excluded because the stereochemistry is not preserved through the pipeline.

For both SPICE and CREST-relaxed SPICE, we align atom ordering between the dataset coordinates and the RDKit molecule built from the canonical SMILES using `GetSubstructMatch`. We split 0.8/0.1/0.1 into train/validation/test. The training set contains 19,198 molecules (959,802 SPICE conformers; 2,203,392 CREST conformers), the validation set 2,508 molecules (125,370; 278,673), and the test set 2,941 molecules.

OE62. OE62 (Stuke et al., 2020) provides PBE-relaxed geometries of crystal-forming organic molecules. We read the SMILES and coordinates, align atom ordering via `GetSubstructMatch` as above, and split 0.8/0.1/0.1. The training set contains 28,038 molecules, the validation set 4,297, and the test set 3,933 (one conformation each).

PDBBind. We use PDBBind-v2020 (Liu et al., 2015). For each protein–ligand pair, we extract the ligand from the SDF file, add hydrogens, and generate a canonical SMILES. Atom ordering between the SDF-derived and SMILES-derived RDKit molecules is aligned via `GetSubstructMatch`.

The protein binding pocket serves as the environment graph. We parse the pocket PDB file and retain only heavy atoms, adding edges between atoms within the same residue. Node features follow ETFlow conventions.

We use the original PDBBind splits. After filtering invalid ligands, the training set has 9,940 examples, the validation set 614, and the test set 218 (one ligand conformation each).

Aquamarine (AQM). Aquamarine (Medrano Sandonas et al., 2024) contains conformations relaxed in implicit water. We build RDKit molecules from SMILES, add hydrogens, and align atom ordering as above. The environment graph uses a fixed RDKit-embedded water proxy with four normalized solvation descriptors appended to node features: cavity surface area (sCAV), cavity volume (vCAV), free energy in electrolyte (eSOLV), and non-electrostatic free energy (eNELEC).

After filtering, we split molecules 0.8/0.1/0.1. The processed files retain all Aquamarine conformers, but the default one-reference protocol selects the dataset's `c1` (index-0) conformer for training and evaluation, yielding 1,276 training, 175 validation, and 157 test molecules. This is a fixed canonical-index choice rather than an energy ranking; we report a conformer-selection sensitivity analysis in Appendix P.

Crystallography Open Database (COD). The COD (Gražulis et al., 2009; 2012) is an open-access collection of crystal structures. We select organic molecular crystals by applying the following filters: the structure must (i) contain organic molecules, (ii) not be marked as disordered (Groom et al., 2016), (iii) not be missing hydrogen, (iv) have a valid SMILES (Weininger, 1988) with a single fragment, (v) pass structural validation (Ong et al., 2013), and (vi) contain exactly one formula unit in the asymmetric unit ($Z' = 1$).

We recover the molecular topology with RDKit maximum-common-substructure matching (`FindMCS`) (Landrum & contributors, 2025) rather than the often-unreliable COD SMILES, and extract the space group, Wyckoff positions, and Cartesian coordinates for each formula unit using `pymatgen` (Ong et al., 2013). We process structures into PyTorch Geometric objects, aligning atom ordering via RDKit `GetSubstructMatch`. Lattice parameters $(a, b, c, \alpha, \beta, \gamma)$ and the space group number are included as features.

After filtering, we split 0.8/0.1/0.1. The training set has 20,765 molecules, the validation set 3,239, and the test set 3,235 (one conformation each).

B Distributions of Atom Counts and Rotatable-Bond Counts

To characterize the intrinsic difficulty of conformer generation across datasets, we report histograms of molecular size (number of atoms, N_{atoms}) and flexibility (number of rotatable bonds, N_{rot}) for each dataset (Figures 6 and 7). We compute N_{rot} using `rdMolDescriptors.CalcNumRotatableBonds` from RDKit with its default definition (non-ring single bonds between non-terminal heavy atoms).

Across the pretraining sets (GEOM Drugs, GEOM QM9, SPICE, CREST-relaxed SPICE, and OE62), N_{atoms} concentrates in the mid-size regime with means in the $\sim$ 17–44 range, while N_{rot} is typically modest (most molecules have fewer than 15 rotatable bonds).

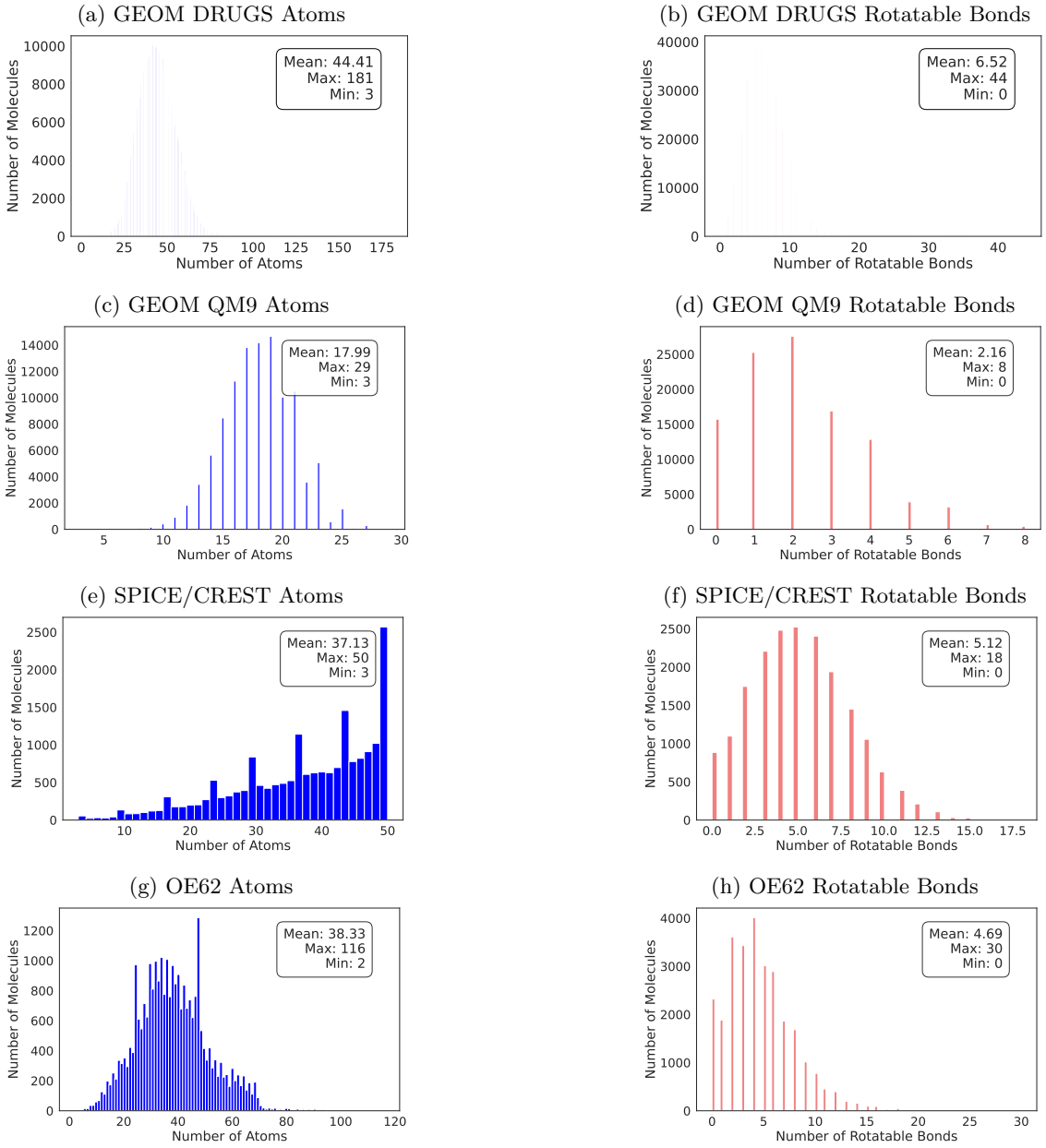


Figure 6: Histograms of atom and rotatable-bond counts for the pretraining datasets: GEOM DRUGS, GEOM QM9, SPICE, CREST-relaxed SPICE, and OE62.

For the finetuning sets (GEOM Drugs, PDBBind, AQM, and COD), the atom-count distribution shifts larger with means in the $\sim 38 - 64$ range. In particular, the PDBBind dataset exhibits a long tail (very large molecules), indicating increased geometric and torsional complexity.

C Training Details and Hyperparameters

All models are trained on 8 NVIDIA A100 GPUs with AdamW and a cosine-annealing learning rate schedule with linear warmup. Table 6 lists the hyperparameters.

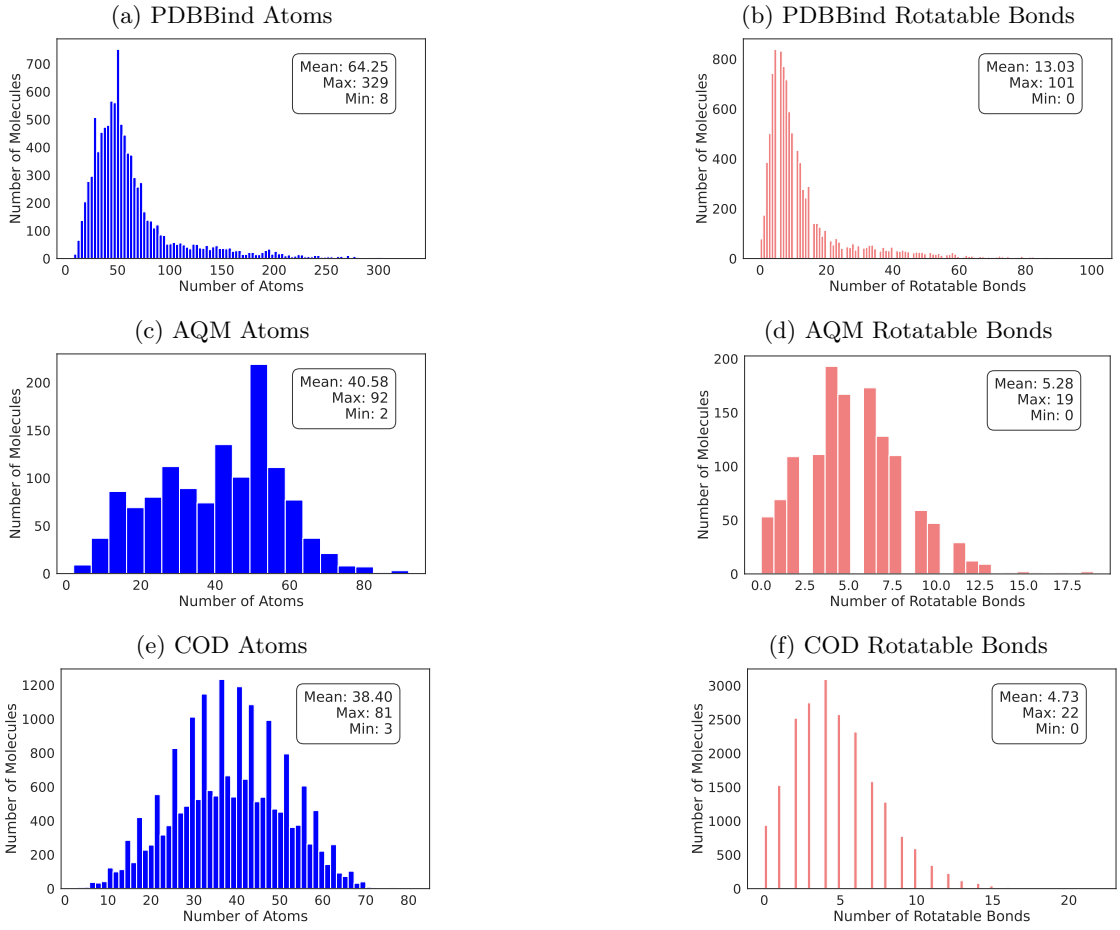


Figure 7: Histograms of atom and rotatable-bond counts for PDBBind, AQM, and COD. GEOM DRUGS is shared with the pretraining datasets (Figure 6).

Pretraining. We train for 800 epochs, resampling 100K examples per epoch from the pretraining pool with mixing probabilities 0.6 (GEOM-DRUGS), 0.2 (OE62), 0.1 (GEOM-QM9), 0.05 (SPICE), and 0.05 (CREST).

Unified finetuning. Starting from the pretrained checkpoint, we finetune for 800 epochs using nominal allocations from a 100K-example epoch: 0.6 (Vacuum / GEOM-DRUGS), 0.2 (Packing / COD), 0.15 (Docking / PDBBind), and 0.05 (Solvation / AQM). Each allocation is capped at the available dataset size, giving 91,216 effective samples per epoch.

Individual finetuning. We finetune on each environment's full training set for 800 epochs using the same optimizer and schedule.

C.1 Training, Inference, and Reproducibility Details

Optimization and schedule. We use AdamW with $(\beta_1, \beta_2) = (0.95, 0.999)$, weight decay 10^{-8}, and gradient-norm clipping at 1.0. The learning rate increases linearly over 31,250 steps to a peak of 5×10^{-4}, then decays along a cosine schedule over a 312,500-step cycle to a minimum of 10^{-8}.

Data splits, validation, and checkpointing. Each environment uses fixed train/validation/test splits. We validate every epoch and monitor the total validation loss (`val/total_loss`); the model checkpoint with the lowest validation loss is used for all reported results.

Table 6: Hyperparameters for EVA-Flow.

Hyperparameter	Value
Batch size	32
FM decoder layers	20
Hidden channels	160
Attention heads	8
Latent dim (d_z)	128
Environment dim (d_e)	128
Encoder layers	3
Base layers	3
Environment layers	3
Cutoff range	[0.0, 10.0] Å
Activation	SiLU
RBF type / count	ExpNorm / 64
Optimizer	AdamW
AdamW (β_1, β_2)	(0.95, 0.999)
Max LR	5×10^{-4}
Min LR	10^{-8}
Weight decay	10^{-8}
Warmup steps	31,250
Cosine cycle length	312,500 steps
Gradient clip (norm)	1.0
Random seed	42
Inference sampler / steps	SDE / 50
SDE noise (s_{churn}, std)	(1.0, 1.0)
GPUs	8× A100
Total parameters	12.1M

Random seeds. We use training seed 42. Inference uses a stochastic SDE sampler. Table 2 reports mean ± standard deviation over five independent sampling seeds from each fixed checkpoint; these values quantify inference-sampler variability, not training uncertainty. The separate all-conformer ETFlow diagnostic uses six runs.

Inference sampler and noise. Unless otherwise noted, all reported numbers use the SDE sampler with 50 integration steps, additive-noise scale std = 1.0, churn $s_{\text{churn}} = 1.0$, and a stochastic window $t \in [10^{-4}, 0.9999]$; the initial state x_0 is drawn from the harmonic prior with no additional perturbation. The deterministic ODE sampler is also supported and trades a small amount of Recall COV for speed (Vacuum: 0.737 ODE vs. 0.747 SDE).

The inference-time perturbation schedule is a sampling heuristic, distinct from the stochastic interpolation $\sigma\sqrt{t(1-t)}$ used to construct training paths. The term "SDE sampler" follows the implementation and does not imply that this procedure is the probability-flow SDE associated with the training interpolation.

Inference time. Table 7 reports per-molecule inference time on a single A100 GPU (GEOM-DRUGS test set). EVA-Flow is modestly faster than retrained ETFlow at the same step count in our implementation; both use their respective graph-conditioned vector fields and the fixed harmonic initialization.

Compute. Models are trained on 8× NVIDIA A100 GPUs. Each 800-epoch stage (pretraining and unified finetuning) takes roughly 5 minutes per epoch, or approximately 67 wall-clock hours (∼530 A100-GPU-hours); individual finetuning is run per environment on its full training set.

Table 7: Per-molecule inference time on a single A100 GPU (GEOM-DRUGS, 1,000 molecules). EVA-Flow uses the Pretrain+Unified setup.

Method	Sampler	Steps	R-COV ↑	Time (s/mol) ↓
ETFlow (retrained)	SDE	50	0.725	0.309
EVA-Flow (P+U)	SDE	50	0.747	0.266
EVA-Flow (P+U)	SDE	100	0.752	0.530

Dataset preprocessing. Each dataset is processed by a dedicated script that will be released in the future. Molecular graphs and features are built with `RDKit` and `datamol`; crystal lattices and space groups for Packing use `pymatgen`, with `OpenBabel` for bond perception from CIF files; numerical processing uses `NumPy` and `SciPy`.

D Evaluation Metric Definition

Following prior work (Ganea et al., 2021; Xu et al., 2022; Jing et al., 2022), we evaluate generated conformers using average minimum RMSD (AMR) and Coverage (COV), each measured for Recall and Precision. Recall measures the extent to which the generated conformers capture the ground-truth conformers, while Precision indicates the proportion of generated conformers that match the ground-truth conformers. Let C_g and C_r denote the generated and reference conformation sets, respectively. The metrics are:

$$\text{AMR-R}(C_g, C_r) = \frac{1}{|C_r|} \sum_{\mathbf{R} \in C_r} \min_{\hat{\mathbf{R}} \in C_g} \text{RMSD}(\mathbf{R}, \hat{\mathbf{R}}) \tag{19}$$

$$\text{COV-R}(C_g, C_r) = \frac{1}{|C_r|} |\{\mathbf{R} \in C_r | \text{RMSD}(\mathbf{R}, \hat{\mathbf{R}}) < \delta, \hat{\mathbf{R}} \in C_g\}| \tag{20}$$

$$\text{AMR-P}(C_r, C_g) = \frac{1}{|C_g|} \sum_{\hat{\mathbf{R}} \in C_g} \min_{\mathbf{R} \in C_r} \text{RMSD}(\hat{\mathbf{R}}, \mathbf{R}) \tag{21}$$

$$\text{COV-P}(C_r, C_g) = \frac{1}{|C_g|} |\{\hat{\mathbf{R}} \in C_g | \text{RMSD}(\hat{\mathbf{R}}, \mathbf{R}) < \delta, \mathbf{R} \in C_r\}| \tag{22}$$

A lower AMR signifies closer matches. Recall COV measures reference coverage, whereas Precision COV measures the fraction of generated samples close to a finite reference set; neither alone establishes sample diversity or calibrated probabilities. The threshold δ is set to 0.5 Å for Solvation, 0.75 Å for Vacuum (Hassan et al., 2024) and Packing, and 2.0 Å for Docking (Corso et al., 2022).

Recall and Precision must be read jointly because Recall grows with sample budget while Precision penalizes samples that do not match the finite reference set.

E Additional Results

E.1 Additional Precision Results

Precision Coverage plots are shown in Figure 8.

E.2 Median Coverage under the 4-Conformer Protocol

Table 8 reports median Coverage. With one reference and four generated conformers per molecule, per-molecule Recall COV is binary and Precision COV is restricted to $\{0, 0.25, 0.5, 0.75, 1\}$. Consequently, the median values are strongly quantized and often identical across models; mean COV and median AMR in Table 2 provide substantially finer discrimination.

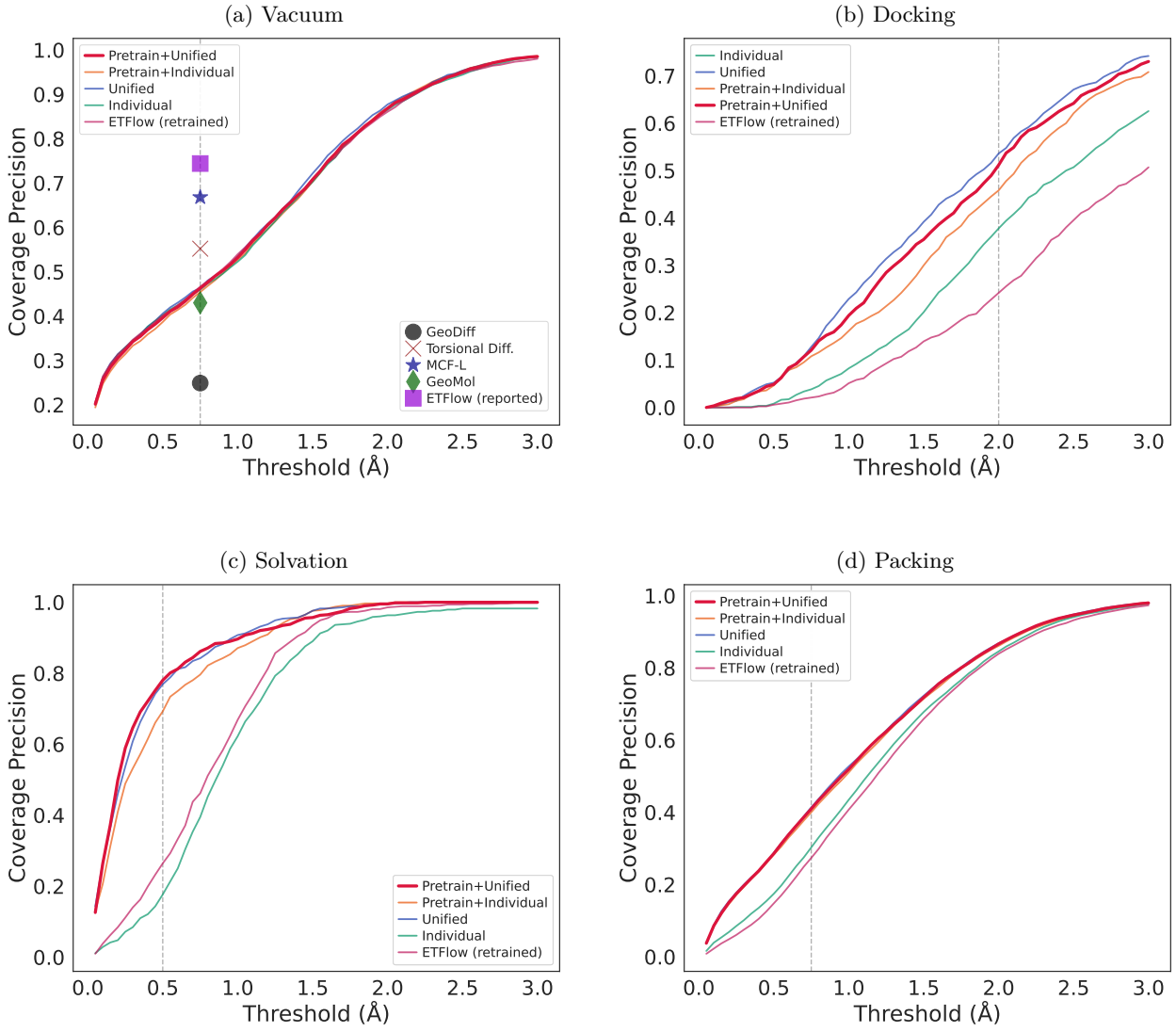


Figure 8: Precision Coverage for all four environments: Vacuum, Docking, Solvation, and Packing.

E.3 Generation Diversity

To test for sample collapse, we measure the mean symmetry-aware heavy-atom best-RMS over all unique pairs of the four generated conformers per molecule, averaged over the test set (Pretrain+Unified model). Table 9 reports non-zero spread in every environment, establishing non-trivial sample diversity across all four settings.

E.4 Multi-Reference Solvation Evaluation

Docking and Packing remain single-reference structure-prediction evaluations. The multi-reference evaluation below concerns Solvation.

Aquamarine test molecules contain a mean of 37 conformers (up to 431), although the default training and 4-conformer evaluation use `c1`. We therefore evaluate against all available references with the $2K$ protocol (Table 10). Recall and Precision are reported jointly: high Recall demonstrates broad coverage of the

Table 8: Median Recall and Precision COV under the 4-conformer protocol.

Environment	Setup	R-COV median ↑	P-COV median ↑
Vacuum	Pretrain+Unified	1.00	0.50
	Pretrain+Individual	1.00	0.50
	Unified	1.00	0.50
	Individual	1.00	0.50
	ETFlow (retrained)	1.00	0.50
Docking	Pretrain+Unified	1.00	0.50
	Pretrain+Individual	1.00	0.25
	Unified	1.00	0.50
	Individual	1.00	0.25
	ETFlow (retrained)	0.00	0.00
Solvation	Pretrain+Unified	1.00	1.00
	Pretrain+Individual	1.00	0.75
	Unified	1.00	1.00
	Individual	0.00	0.00
	ETFlow (retrained)	1.00	0.25
Packing	Pretrain+Unified	1.00	0.25
	Pretrain+Individual	1.00	0.25
	Unified	1.00	0.25
	Individual	1.00	0.25
	ETFlow (retrained)	0.00	0.00

Table 9: Pairwise-RMSD diversity (Å) among the four generated conformers per molecule (Pretrain+Unified).

Environment	Mean ↑	Median	Std
Vacuum	0.915	0.903	0.620
Docking	2.227	1.828	1.508
Solvation	0.456	0.359	0.315
Packing	0.967	0.912	0.634

available ensemble, while Precision penalizes generations that do not match that finite reference set. The high Recall (0.981) shows the single-conformer-trained model covers the available solvated ensemble.

E.5 Physical Reference Methods

We add two non-learned reference methods under the same four-sample evaluation budget: RDKit ETKDG followed by MMFF relaxation for all environments, and GFN2-xTB with ALPB-water implicit solvation for Solvation. These are physical reference points rather than controlled ablations of environment conditioning because they differ from EVA-Flow in model class, data, and optimization.

Table 10: Solvation evaluation with single- and multi-reference protocols.

Training	Protocol	Refs/mol	Samples/mol	R-COV ↑	R-AMR ↓	P-COV ↑	P-AMR ↓
Single conformer	4-conf.	1	4	0.936	0.157	0.693	0.431
Single conformer	2K	mean 37	$2K$	0.981	0.102	0.541	0.591
Multiple conformers	2K	mean 37	$2K$	0.962	0.111	0.518	0.612

Table 11: Non-learned reference methods under the 4-conformer protocol.

Environment	Method	R-COV ↑	R-AMR ↓
Vacuum	ETKDG+MMFF	0.409	1.18
	EVA-Flow (P+U)	0.747	0.468
Docking	ETKDG+MMFF	0.581	2.09
	EVA-Flow (P+U)	0.684	1.899
Solvation	ETKDG+MMFF	0.006	3.42
	GFN2-xTB+ALPB-water	0.006	3.38
	EVA-Flow (P+U)	0.949	0.153
Packing	ETKDG+MMFF	0.606	0.72
	EVA-Flow (P+U)	0.644	0.664

E.6 Reconciling Published and Retrained ETFlow Results

Baseline scope and training. For Solvation and Packing, no established learned baselines share our task definitions and splits; we report retrained ETFlow under identical evaluation. For Docking, retrained ETFlow is selected by validation loss after 100 epochs on the same PDBBind split, molecular inputs, sample budget, and evaluator. ETFlow has no published internal-conformation result under this protocol. This is a matched ligand-only reference, but a shared schedule does not guarantee architecture-specific optimal tuning; the inference-time NoPocket ablation (Appendix H) more directly tests whether EVA-Flow uses pocket information.

Vacuum comparisons. On Vacuum under the 2K protocol (Table 3), EVA-Flow is comparable to retrained ETFlow (Recall COV 0.721 vs. 0.712; Precision COV 0.707 vs. 0.704) and remains behind the 20×-larger MCF-L on Recall (0.721 vs. 0.847; 12.1M vs. 242M parameters), while exceeding MCF-L on Precision COV (0.707 vs. 0.668). EVA-Flow is not a new state of the art for Vacuum.

The ETFlow number reported in its paper uses the original GEOM-DRUGS benchmark, all available reference conformers, and a $2K$ sampling budget. This differs from our default Table 2, which uses the chemically cleaned benchmark, one reference, and four samples. Table 12 evaluates the published checkpoint and retrained ETFlow/EVA-Flow models through our evaluation code. The original and cleaned test sets share 998/1,000 molecules, but differ in preprocessing and reference counts (up to 2,497 versus a cap of 30), which also changes the $2K$ sample budget. Comparisons within a matched block support model comparisons; values across datasets or implementation pipelines are diagnostic only and are not used for causal attribution.

F Physical Validity of Generated Conformers

Coverage and AMR measure geometric similarity to reference conformers but not energetic plausibility. To assess physical validity directly, we relax all 4,000 generated Vacuum conformers (1,000 molecules $\times$ 4 conformers) with UMA (Wood et al., 2026), a large-scale pretrained molecular potential, and measure the energy change and geometric shift upon relaxation (Table 13). The very small energy changes and atomic displacements indicate that EVA-Flow generates conformers near local energy minima under UMA.

Table 12: Matched $2K$ evaluation (0.75 Å) on original and cleaned GEOM-DRUGS test sets. All rows within a block use identical evaluation code and test references.

Model	Training set	Test set	R-COV ↑	R-AMR ↓	P-COV ↑	P-AMR ↓
Published ETFlow checkpoint	Original	Original	0.549	0.766	0.339	1.057
ETFlow (retrained)	Original	Original	0.712	0.562	0.704	0.594
EVA-Flow	Original	Original	**0.721**	**0.550**	**0.707**	**0.589**
Published ETFlow checkpoint	Original	Cleaned	0.517	0.799	0.294	1.193
ETFlow (retrained)	Original	Cleaned	0.759	0.505	0.629	0.688
EVA-Flow	Original	Cleaned	**0.763**	**0.503**	**0.632**	**0.686**
ETFlow (retrained)	Cleaned	Cleaned	0.798	0.443	0.728	0.532
EVA-Flow	Cleaned	Cleaned	**0.810**	**0.415**	**0.740**	**0.505**

Table 13: UMA relaxation of EVA-Flow Vacuum conformers (4,000 conformers from 1,000 molecules).

Metric	Value
UMA relaxation converged	99.9% (3,995/4,000)
Mean ΔE after relaxation	0.302 kcal/mol
Median ΔE after relaxation	0.115 kcal/mol
Fraction with $\Delta E < 1$ kcal/mol	97.1%
Mean RMSD shift during relaxation	0.056 Å
Fraction with RMSD shift < 0.1 Å	84.2%

Table 14: Shared-molecule analysis after excluding all shared molecules from finetuning.

Pair (A/B)	n	own A	own B	cross A	cross B	generation shift
Solvation–Vacuum	142	0.29	0.90	3.35	3.25	3.32
Docking–Vacuum	558	0.95	0.81	1.17	1.19	1.16
Vacuum–Packing	487	0.38	0.43	0.64	0.59	0.62
Docking–Solvation	13	0.63	0.35	2.34	2.27	2.23
Packing–Solvation	23	0.14	0.14	1.48	1.47	1.50
Packing–Docking	64	0.36	0.46	0.51	0.37	0.40

G Shared-Molecule Controls

Shared molecules are matched by canonical SMILES and aligned with consistent heavy-atom mappings before RMSD calculation. Table 14 reports the control in which all 1,201 molecules occurring in at least two finetuning environments are removed from every finetuning training set. For pair A/B, own A/B compares generations conditioned on each context with that context's reference; cross A/B swaps the reference; and generation shift compares the two context-conditioned generations. All entries are mean heavy-atom RMSD in Å. Because the broad pretraining corpus includes Vacuum molecules, this control isolates finetuning exposure but does not guarantee pretraining decontamination.

The own-versus-cross pattern remains for pairs with well-separated references, while Packing–Docking stays weakly separated because those reference geometries are physically similar. These held-out data show the conditioning effect persists on molecules never seen in training, ruling out finetuning memorization as the explanation.

Table 15: Environment-feature sensitivity (inference-time zeroing, no retraining). Thresholds: Solvation 0.5 Å, Packing 0.75 Å, Docking 2.0 Å.

Environment	Ablation	R-COV ↑	R-AMR ↓	P-COV ↑	P-AMR ↓
Solvation	Full	**0.949**	**0.153**	**0.780**	**0.367**
	NoSolvDesc	0.905	0.191	0.734	0.399
Packing	Full	**0.644**	**0.664**	**0.410**	**1.073**
	NoLattice	0.575	0.762	0.360	1.154
Docking	Full	**0.684**	**1.899**	**0.512**	**2.471**
	NoPocket	0.400	2.590	0.257	3.197

H Environment-Graph Sensitivity

We measure the performance change after zeroing domain-specific environment features at inference time (Table 15). **NoSolvDesc** zeros the four solvation descriptors; **NoLattice** zeros the seven lattice/space-group parameters; **NoPocket** zeros the entire environment embedding for docking. All three ablations degrade performance, confirming that the model relies on each environment component, and the docking pocket is the most critical.

I Full Ablation Metrics

Replacing the FM decoder with regression targets the conditional mean of a multi-modal distribution, explaining the collapse in Recall COV for NoFM.

Figure 5 reports Recall COV for the original ablations. Table 16 provides their complete metrics across all four environments. NoBase reduces performance within the original base–latent parameterization; the full latent-free training-regime grid (Appendix J) shows that removing both the learned base and latent (the latent-free variant) has mixed effects under Individual training but trails the full model in all twelve cross-environment transfer comparisons.

J Latent-Free Model: Full Training-Regime Grid

To directly test whether the base and latent are necessary, we train the latent-free variant (environment embedding + fixed harmonic base + FM decoder; both the learned base and the VAE latent removed) in all four training regimes and evaluate every environment. Table 18 reports Recall COV for the latent-free model against the full model (Table 2) in each cell. Under Individual training, full-model to latent-free Recall COV changes from 0.728 to 0.727 in Vacuum, 0.553 to 0.470 in Docking, 0.414 to 0.503 in Solvation, and 0.512 to 0.493 in Packing. Thus, performance is nearly unchanged in Vacuum, improves in Solvation, and declines in Docking and Packing. In contrast, the latent-free model trails in all three cross-environment transfer regimes (Unified and the two pretraining-based settings): every one of the twelve transfer cells favors the full model, with the gap concentrated in the low-data Solvation environment (-0.16 to -0.26 Recall COV, several times the sampler standard deviation). These results support a consistent benefit from the coupled base and latent in the tested cross-environment transfer regimes; their contribution under Individual training depends on the environment.

Table 16: Full ablation metrics (Unified setup, small model). Thresholds: Vacuum/Packing 0.75 Å, Solvation 0.5 Å, Docking 2.0 Å. Best per metric/environment in **bold**.

Metric	Setup	Vacuum	Docking	Solvation	Packing
R-COV ↑	EVA (Full)	**0.740**	**0.674**	**0.930**	**0.640**
	GCN	0.670	0.665	0.803	0.605
	Gaussian	0.405	0.237	0.822	0.312
	NoConsist	0.338	0.577	0.134	0.373
	NoBase	0.321	0.577	0.255	0.398
	NoFM	0.000	0.126	0.006	0.005
P-COV ↑	EVA (Full)	**0.463**	**0.536**	**0.769**	**0.414**
	GCN	0.396	0.490	0.497	0.384
	Gaussian	0.241	0.179	0.618	0.201
	NoConsist	0.184	0.379	0.102	0.206
	NoBase	0.177	0.376	0.140	0.220
	NoFM	0.000	0.057	0.003	0.002
R-AMR ↓	EVA (Full)	**0.456**	**1.889**	**0.166**	**0.682**
	GCN	0.623	2.153	0.285	0.714
	Gaussian	1.078	4.528	0.291	1.248
	NoConsist	1.108	2.221	0.896	0.990
	NoBase	1.113	2.160	0.790	0.963
	NoFM	2.136	3.536	1.735	2.214
P-AMR ↓	EVA (Full)	**0.954**	**2.391**	**0.370**	**1.067**
	GCN	1.117	2.937	0.593	1.117
	Gaussian	1.551	5.414	0.492	1.601
	NoConsist	1.569	2.941	0.921	1.400
	NoBase	1.560	2.711	0.902	1.363
	NoFM	2.502	4.020	2.145	2.581

Table 17: Full architecture (reported model): Recall COV at the environment threshold across all four training regimes.

Environment	Individual	Unified	Pretrain+Ind.	Pretrain+Unified
Vacuum	0.728	0.740	0.733	0.747
Docking	0.553	0.674	0.656	0.684
Solvation	0.414	0.930	0.936	0.949
Packing	0.512	0.640	0.633	0.644

Table 18: Latent-free model (learned base and latent removed): Recall COV at the environment threshold across all four training regimes. Compared with the full model (Table 17), Individual performance is nearly unchanged in Vacuum, higher in Solvation, and lower in Docking and Packing. All twelve transfer comparisons favor the full model, with the largest gaps in Solvation.

Environment	Individual	Unified	Pretrain+Ind.	Pretrain+Unified
Vacuum	0.727	0.708	0.688	0.672
Docking	0.470	0.660	0.614	0.628
Solvation	0.503	0.675	0.771	0.771
Packing	0.493	0.622	0.615	0.613

Table 19: Top: total molecule exposure across training. Bottom: balanced vs. default sampling (Recall COV at the environment threshold).

Environment	Dataset	Unified/ep	Unified tot.	Indiv./ep	Indiv. tot.
Vacuum (DRUGS)	272,641	60,000	48.0M	272,641	218.1M
Packing (COD)	20,765	20,000	16.0M	20,765	16.6M
Solvation (AQM)	1,276	1,276	1.02M	1,276	1.02M
Docking (PDB)	9,940	9,940	7.95M	9,940	7.95M

Environment	Default (60/20/5/15)	Balanced (25/25/25/25)	Change
Vacuum (0.75 Å)	0.740	0.692	−6.5%
Docking (2.0 Å)	0.684	0.670	−2.0%
Solvation (0.5 Å)	0.949	0.847	−10.7%
Packing (0.75 Å)	0.644	0.643	−0.2%

K Data Exposure and Balanced Sampling

To test whether the Unified advantage stems from cross-environment transfer rather than data volume, we analyze total molecule exposure and run a balanced-sampling control. Table 19 (top) counts molecule samples over the stated 800 epochs. Unified starts from nominal 60/20/5/15% allocations of 100K samples, then caps each environment at its dataset size, yielding 91,216 effective samples per epoch; Individual models pass over their full dataset each epoch. Unified sees far fewer DRUGS samples than Individual (48.0M vs. 218.1M) yet outperforms it; for the small-data environments the Individual models already overfit (validation loss plateaus by epoch ∼30–50), so additional repetitions alone cannot explain the gains.

Table 19 (bottom) compares balanced (25/25/25/25%) against the default weighted sampling, both from the same pretrained checkpoint. Solvation and Docking see the *same* number of molecules per epoch under both schemes (capped at dataset size), yet Solvation drops 10.7% when the DRUGS allocation falls from 60% to 25%, indicating that DRUGS data benefits Solvation through cross-environment transfer in the shared backbone, not through Solvation data volume.

L Train–Test Consistency of the Base Distribution

At inference the initial coordinates are drawn from the simple harmonic prior rather than the learned training-time base. The measured divergence at convergence is 0.036 nats (validation; 0.026 train). On 200 test molecules, the learned-to-prior eigenvalue ratio has median 1.0000 and all measured modes are within 1% of unity, with posterior mean approximately zero. A noise-injection sweep also degrades gradually (Table 20). These diagnostics indicate that the inference-time base mismatch is small. The large NoBase drop in the original parameterization is therefore an optimization interaction, not an inference-time mismatch: its latent activates while the full-model latent collapses (Table 21), whereas the effect of the latent-free model under Individual training varies by environment (Table 18). The base and latent are thus essential for low-data cross-environment transfer while remaining inactive at inference.

Table 20: Inference noise-injection sweep on Vacuum (Pretrain+Unified checkpoint, SDE 50 steps): perturbing the harmonic prior by $\epsilon \cdot \mathcal{N}(\mathbf{0}, I)$.

ϵ	R-COV ↑	R-AMR ↓	P-COV ↑	P-AMR ↓
0.00 (default)	**0.747**	**0.468**	**0.464**	**0.979**
0.01	0.738	0.470	0.455	0.983
0.05	0.735	0.493	0.459	0.981
0.10	0.745	0.471	0.464	0.973
0.20	0.730	0.486	0.452	0.985
0.50	0.734	0.480	0.453	0.982
1.00	0.704	0.529	0.413	1.079

Table 21: Latent activation in the full and NoBase models.

Model	Active dims	KL/node (nats)	Pooled $\|\mu\|$	Environment-dependent
Full	0/128	$\approx 10^{-6}$	$\approx 10^{-3}$	No (cosine 1.000)
NoBase	3/128	up to ~ 3	≈ 0.21	Yes

Table 22: Latent-collapse diagnostics and intervention. "Random" samples $z \sim \mathcal{N}(\mathbf{0}, I)$; "zero" replaces the matched latent draw by zero. COV values use the environment-specific thresholds.

Environment	KL/node mean	KL/node median	Active dims	Posterior $\|\mu\|$	R-COV random	R-COV zero	P-COV random	P-COV zero
Vacuum	1.5×10^{-6}	1.4×10^{-6}	0/128	8.2×10^{-5}	0.707	0.706	0.433	0.433
Docking	0.0476	1.4×10^{-6}	0/128	1.9×10^{-4}	0.661	0.661	0.497	0.497
Solvation	1.6×10^{-6}	1.5×10^{-6}	0/128	8.2×10^{-5}	0.803	0.809	0.580	0.591
Packing	1.6×10^{-6}	1.5×10^{-6}	0/128	8.2×10^{-5}	0.649	0.649	0.419	0.418

M Latent Posterior-Collapse Diagnostics

We diagnose whether the variational latent z carries information in the Pretrain+Unified checkpoint. We compute the per-node latent KL, posterior statistics, and active dimensions (a dimension is active when the variance across data of its posterior mean exceeds 10^{-2}), then rerun generation after replacing sampled z by zero while preserving all subsequent random draws. Table 22 reports the paired random-versus-zero intervention using one matched diagnostic sampling seed, rather than the five-seed means in Table 2. The posterior is effectively collapsed: Vacuum, Packing, and Solvation have mean KL approximately 1.5×10^{-6} nats per node; Docking has a larger mean due to a small tail but the same near-zero median; no dimensions are active. For the same molecule across all six environment pairs, the pooled posterior mean has norm approximately 10^{-3} and is unchanged across contexts (cosine 1.000, relative $L_2 \leq 0.001$). Setting $z = 0$ leaves all metrics effectively unchanged. Thus the decoder primarily conditions on G and e, and we do not attribute its performance to information carried by z. The NoConsist degradation is instead interpreted as an optimization effect of that objective; the latent-free variant shows that z can be removed along with the learned base.

N Loss-Weight Sensitivity

We use unit weights because the implemented losses are normalized per node/batch and have comparable scale.

We vary one regularizer weight at a time by $0.1\times$ and $10\times$, keeping the FM weight and training schedule fixed, and report the Recall-COV change from the unit-weight control (Table 23). In 21/24 cells, $|\Delta| < 0.045$. The larger changes are Vacuum with latent-KL $10\times$ (-0.045), Solvation with consistency $0.1\times$ (-0.064), and Solvation with latent-KL $10\times$ (-0.141). Thus most cells are stable to large one-at-a-time changes, while

Table 23: One-factor-at-a-time loss sensitivity: change in Recall COV from unit weights.

Weight change	Vacuum	Docking	Solvation	Packing
latent-KL 0.1×	−0.018	−0.010	−0.020	−0.007
latent-KL 10×	−0.045	+0.018	−0.141	−0.030
base-KL 0.1×	−0.027	+0.004	+0.019	−0.008
base-KL 10×	−0.014	+0.032	+0.019	−0.015
consistency 0.1×	−0.018	+0.009	−0.064	−0.010
consistency 10×	−0.004	+0.028	+0.025	−0.001

Table 24: Equivariant environment encoder minus matched GCN control. Positive ΔR-COV and negative ΔR-AMR favor the equivariant encoder.

Environment	Coordinate content of G_E	ΔR-COV	ΔR-AMR
Vacuum	zero-coordinate scaffold	−0.003	−0.011
Docking	physical pocket coordinates	+0.028	−0.082
Solvation	fixed proxy; scalar descriptors	−0.153	+0.139
Packing	RDKit proxy and lattice metadata	−0.006	+0.015

Solvation is sensitive to particular latent-regularization settings. This experiment does not vary network capacity or interactions among weights.

O Environment-Encoder Equivariance

We chose the GCN environment encoder as a common interface because some contexts have physical coordinates (pockets and molecular/crystal proxies) whereas others are descriptor-only; for Docking, we jointly rotate ligand and pocket during training to reduce frame dependence.

We replace the GCN environment encoder with a matched SE(3)-equivariant TorchMD encoder under the same controlled Unified training budget (Table 24). Accuracy improves for Docking, whose context carries molecule-specific coordinates, while the encoder is neutral or worse for contexts dominated by topology, proxies, or scalar descriptors. This is an encoder comparison, not a claim that one architecture is uniformly superior.

For a separate joint rigid-rotation stress test over all 218 Docking test complexes, the GCN environment embedding changes by 1.33% on average (0.98% median relative error), with mean cosine similarity 0.99998 (minimum 0.99983). The equivariant encoder changes by approximately 10^{-5} and has cosine similarity 1.0000. This measures embedding stability, not end-to-end equivariance of generated coordinates.

P Aquamarine Conformer-Selection Sensitivity

The default protocol selects the stored `c1` conformer (array index 0), not an energy-ranked ground state. To quantify reference sensitivity, we hold the trained model, molecule split, sampling seed, and four-conformer budget fixed while changing the test reference to either the minimum-`eSOLV` conformer or one deterministic random conformer per molecule (Table 25). We separately train matched 200-epoch models with each target rule (Table 26). These controlled runs use a smaller budget than the 800-epoch model and are compared only within their block.

Q Molecule Visualization

Table 27 visualizes reference and generated conformers for molecules shared between Packing and Docking.

Table 25: Aquamarine test-reference sensitivity for a fixed model and sampling seed (0.5 Å threshold).

Reference selection	R-COV ↑	R-AMR ↓	P-COV ↑	P-AMR ↓
`c1` (default)	0.803	0.291	0.580	0.538
Minimum `eSOLV`	0.777	0.319	0.588	0.538
Deterministic random	0.803	0.292	0.584	0.538

Table 26: Aquamarine training-target sensitivity in matched 200-epoch runs.

Training-target selection	R-COV ↑
`c1` (default rule)	0.580
Minimum `eSOLV`	0.561
Deterministic random	0.618

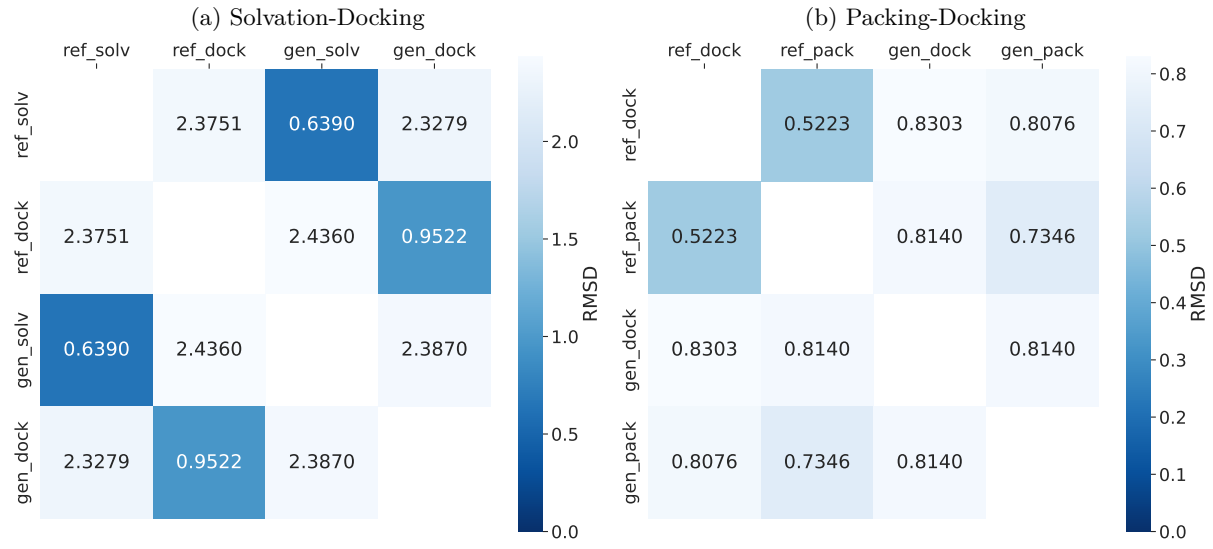


Figure 9: Pairwise heavy-atom RMSD (Å) between reference and generated conformers in the original overlap sets: (a) Solvation-Docking ($n = 12$), and (b) Packing-Docking ($n = 59$). In each heatmap we report the mean RMSD after alignment for: ref_A ↔ ref_B, gen_A ↔ ref_A, gen_A ↔ ref_B, gen_B ↔ ref_A, gen_B ↔ ref_B, and gen_A ↔ gen_B (A/B = the two tasks in the panel). Darker cells indicate lower RMSD (better agreement).

Table 27: Visualization of reference and generated conformations for the shared molecules in Packing-Docking. The molecules are aligned to the Docking reference conformers.

Packing-Docking			
ref_dock	gen_dock	ref_pack	gen_pack
